SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I G. Amrutha (192511232) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha.

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

Cloud architecture for a video streaming platform.

(A) proposed Architecture

the media company should use a scalable cloud architecture with content delivery networks (CDNs), load balancing, auto-scaling, and caching.

millions of concurrent users

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DNS and Global Traffic management

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CDN and Edge Caching

(Delivery videos by nearby location)

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Load Balancer

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Auto-Scaling Application servers

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Cache — Redis/Memcached

Storage — object storage for videos.

(B) Scalability and caching techniques.

Horizontal scaling, CDN, caching, object storage, Adaptive bitrate streaming.

(C) Load balancing and Auto-Scaling.

Load balancing, Auto-Scaling, Health checks. These techniques provide high availability, faster content delivery smooth streaming.

(D) Cost optimization strategies

1) Use CDN caching to reduce repeated requests to origin servers.

2) Use auto-scaling to avoid paying for unused content computing resources.

Conclusion:

CDN, caching, load balancing, and auto-scaling help deliver videos efficiently to millions of users while controlling costs.

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2) Container orchestration and multi-cloud deployment for Fintech.

(A) Proposed solution

The fintech startup should use Kubernetes-based container orchestration to deploy and manage applications consistently across multiple cloud providers.

CI/CD pipeline

Build, test, scan, and deploy containers.

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Container Registry

versioned, signed container images.

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Cloud provider A
Kubernetes cluster
Application pods

Cloud provider B
Kubernetes cluster
Application pods.

(B) Container orchestration.

Kubernetes manages containers automatically by:

=> Scheduling containers on available nodes.

=> Restarting failed containers

=> maintaining the desired number of replicas.

=> Performing rolling updates and rollbacks.

(C) Servic discovery and Scaling.

Servic discovery, Load balancing Horizontal Pod Autoscaler, cluster autoscaling.

(D) Benefits of multi-cloud

1) Reduces dependance on a single cloud provider.

2) Improves resilience against provider-specific failures

(E) Risks and challenges

1) Increased management complexity.

2) Different cloud Pricing models and networking systems.

3) Difficult data synchronization and consistency management.

4) Higher operational costs and staff training requirements.

3. Backup and disaster Recovery
Solution: Use automated backups, data replica and file versioning.
Backup: Store copies in a separate region or account.
RPO: maximum acceptable amount of data loss measured in time.
Business Continuity: Detect failures, restore data, switch to backup systems, and test recovery regularly

4. Hybrid cloud for Government
Architecture: Keep sensitive applications on-premises and non-sensitive applications in the public cloud.
Secure Communication: Use VPN, Firewalls, and private connections.
Identity management: Implement MFA, role-based access control, and centralized authentication.
Encryption: Encrypt data both at rest and in transit.
Challenges: Complex management, network latency, security risks, and Compliance requirements

Multi- Region High Availability for SaaS.

Architecture : Deploy applications in two or more cloud regions.

Replication : Synchronize databases and storage across regions.

Failover : Automatically redirect traffic to a healthy region when one fails.

Monitoring : Track server health, latency, errors and replication delays.

Alerting : Notify administrators about failures and unusual activity.

Benefits : High availability, fault tolerance, and reduced downtime.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented